SAMPLE 3 OF 4

VeamX Data Platform -- AI-Assisted Anomaly Detection & Ransomware Safety Card

Platform Ref	DR-9 Series -- VeamX Data Platform v12, AI/ML Anomaly Detection Module . Documentation Cycle: Q2-Q3 2023
Card Version	v1.0 . Pre-General Availability Safety Disclosure
Author	E. Hannan -- Documentation Lead; surfaced undisclosed adversarial evasion gap during review cycle (see Section 4.3)
Source Team	VeamX Product Management . ML Engineering . External Threat Intel Partner . Customer Success (Healthcare Vertical)
Audience	CISOs . Backup Administrators . Healthcare IT Compliance . Cyber Insurance Underwriters
Classification	Pre-GA Restricted -- NDA Protected . Distribution limited to named recipients only
Related Artifacts	VeamX Security Bulletin v12 . HIPAA Compliance Annex . Ransomware Recovery SLA Template
Software	VeamX Data Platform v12 (GA: February 2023) . Backup & Replication v12 . ML anomaly detection module (entropy + behavioral analysis)

1. Executive Summary

VeamX Data Platform v12 introduces an AI/ML anomaly detection module designed to identify ransomware encryption events, unauthorized bulk deletion activity, and backup policy violations in real time -- before they propagate to the point of unrecoverable data loss. This card documents the module's capabilities, evaluated performance against known and unknown threat families, known limitations, and the rationale behind the pre-GA deployment scope decision.

The pre-GA release is restricted to a cohort of twelve enterprise customers in the healthcare and financial services verticals, selected for advanced security operations maturity. This card exists because customers deploying AI safety systems to protect their most critical data have a right to know exactly what the system can and cannot do before committing to production deployment. We do not soften findings for commercial reasons; the credibility of this document depends on not doing so.

2. AI/ML Module Capabilities

2.1 Detection Performance Summary

Evaluation Metric	Result	Documentation Note
Ransomware detection -- in-distribution	94.7%	Validated on known ransomware families active as of Q1 2023: Conti, LockBit (variants 2.0 and 3.0), BlackCat/ALPHV; rule-based component supplements ML layer
Ransomware detection -- out-of-distribution	71.3%	Novel and unknown variant families; 23.4-point gap is the central safety finding in this card. OOD performance, not in-distribution, is the operationally relevant figure.
False positive rate -- in-distribution	0.8%	Backup job interruptions per 1,000 backup events; measured across pre-GA cohort over 90 days
False positive rate -- out-of-distribution	3.1%	Elevated in environments with high-entropy legitimate I/O: genomics processing, video rendering, database compaction
Bulk deletion detection	98.2%	Rule-based component; near-ceiling performance; not ML-dependent for this detection class
Mean time to alert	Below 45 seconds	From anomaly onset signature to SIEM notification; measured at median across pre-GA cohort
Adversarial evasion resistance	Not evaluated	Critical known unknown -- see Section 4.3. This finding was surfaced during documentation review, not engineering review.

2.2 Validated Deployment Environments (v12 GA, February 2023)

- VMware vSphere 7.x / 8.x (primary validated hypervisor platform; largest share of pre-GA cohort)
- Microsoft Hyper-V 2019 and 2022
- Physical Windows Server environments via VeamX Agent for Windows v6
- AWS, Azure, Google Cloud (cloud-native backup workloads via VeamX Cloud Connect v12)

3. Safety Architecture

The anomaly detection module operates in a read-only observational posture: it monitors backup job telemetry, storage I/O patterns, and file entropy signals without direct access to production data volumes. This was a deliberate architectural choice -- an AI safety component should not create new attack surface on the systems it is designed to protect.

- Inference runs on isolated compute within the VeamX Backup & Replication server; no telemetry is transmitted to external endpoints during detection
- Model updates are cryptographically signed and signature-verified before installation; automatic model updates are disabled in production deployment mode by default

- All detection events are written to an immutable audit log; the logging infrastructure is network-isolated from the environments under monitoring
- Human confirmation is required before any automated remediation action (backup job pause, quarantine snapshot initiation); the AI component produces recommendations, not autonomous actions

4. Known Limitations & Critical Disclosures

4.1 Out-of-Distribution Detection Gap (Priority: High)

The 23.4-point gap between in-distribution and OOD detection performance is the central safety finding in this card and the primary reason the pre-GA release is restricted to twelve customers. Novel ransomware families -- which represent the actual threat landscape for production deployments -- are systematically harder to detect than variants in the training set. This is not a bug; it is a fundamental property of supervised ML systems trained on historical threat data. Customers should treat the 94.7% figure as the ceiling for known threat families and calibrate their incident response posture for the OOD case, which is where a sophisticated attacker operates.

4.2 Elevated False Positive Rate in High-I/O Environments (Priority: Medium)

Environments with unusually high but legitimate entropy in their I/O patterns -- genomics data processing, high-throughput video rendering pipelines, large-scale database compaction jobs -- generate false positive rates approaching the OOD rate of 3.1%. Three of the twelve pre-GA customers operate environments of this type. We briefed each individually and they are operating with adjusted alert thresholds and mandatory human review of all ML-triggered alerts until detection calibration data from their specific I/O profiles is incorporated into a subsequent model update.

4.3 Adversarial Evasion: A Known Unknown (Author Escalation)

DOCUMENTATION LEAD ESCALATION	This finding was surfaced during the documentation review cycle, not by the engineering team. The ML Engineering team had not evaluated whether the anomaly detection model could be evaded by a sophisticated adversary who understood its detection logic. Academic literature on adversarial ML published between 2021 and 2023 demonstrates clearly that entropy-based detection systems are vulnerable to entropy-normalized encryption techniques that preserve the statistical profile of benign I/O patterns while still encrypting file contents.

After escalation, the VeamX product team acknowledged the gap and added the following language to all pre-GA customer agreements: 'The anomaly detection module has not been evaluated against adversarial evasion by a sophisticated threat actor with knowledge of the detection methodology. Customers in elevated-risk threat environments should treat this component as one layer within a defense-in-depth strategy, not as a primary control.' This card documents the escalation in full because surfacing an undisclosed limitation during documentation review -- against commercial pressure to proceed without it -- is exactly the accountability function that safety documentation must perform.

5. Healthcare Equity & Population Impact

Data protection has equity dimensions that are rarely documented at this level of specificity, and we include them here because they belong. Three of the twelve pre-GA customers are regional healthcare systems serving predominantly low-income and rural patient populations -- communities where a successful ransomware attack against the health system results in documented patient harm because there is no alternative care facility within reach. Rural hospitals temporarily closed by ransomware attacks in 2022 and 2023 produced measurable increases in patient transfer times and adverse outcomes in regional trauma cases. The AI safety system documented in this card carries real stakes for populations who will never read this card. That weight should be visible in how we produce it.

6. Deployment Decision Rationale

Restricting the pre-GA release to twelve customers with advanced security operations maturity reflects a proportional release philosophy. The documented OOD detection gap and the unresolved adversarial evasion evaluation create a risk profile that is manageable for organizations with mature incident response capabilities and potentially hazardous for organizations that would rely on this module as a primary ransomware control without that maturity. We document this rationale here explicitly because deployment scope decisions -- not only capability claims -- require accountability documentation.